

Highlights

Physics-informed LSTM and BiGRU-based approach for maneuverable satellite cluster trajectory prediction and intention recognition

aaa, bbb, ccc

- Physics-informed deep learning framework for satellite trajectory prediction
- BiGRU-based architecture for maneuver intention recognition
- Integration of orbital dynamics constraints with data-driven models

Physics-informed LSTM and BiGRU-based approach for maneuverable satellite cluster trajectory prediction and intention recognition^{*}

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ABSTRACT

This paper proposes a novel approach combining physics-informed long short-term memory (LSTM) networks and bidirectional gated recurrent units (BiGRU) for maneuverable satellite cluster trajectory prediction and intention recognition. The physics-informed framework incorporates orbital dynamics constraints into the learning process, while the BiGRU-based architecture captures temporal dependencies from both forward and backward directions for intention recognition. The proposed method is validated through comprehensive numerical simulations.

1. Introduction

2. Problem statement

3. Physics-informed LSTM trajectory prediction

4. BiGRU-based intention recognition

5. Simulations and analysis

6. Conclusions

CRedit authorship contribution statement

aaa: Conceptualization, Methodology, Software, Writing – original draft. **bbb:** Formal analysis, Validation, Writing – review & editing. **ccc:** Data curation, Investigation, Visualization.

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